



FP1 Chapter 4 Polar Coordinates

Lecture 4 Polar Coordinates

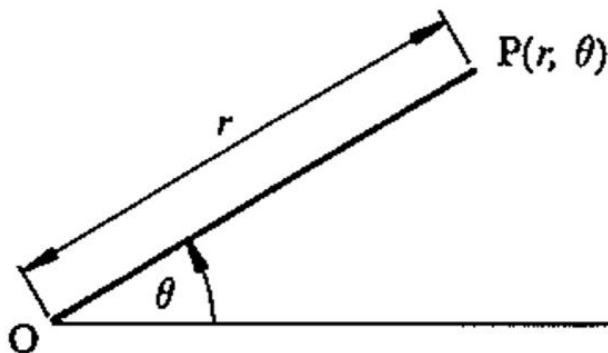
(Rectangular vs Polar Coordinates, Polar Curves, Area of Sector)

Polar Coordinates

- Cartesian coordinates are the most used system for fixing the position of points in a plane but some curves have very complicated Cartesian equations and, in such cases, a different coordinate system may be more useful.
- Polar coordinates are another alternative system which we may find it useful when studying certain curves.

Polar Frame of Reference:

- This system of reference consists of a fixed point O , called the **pole** and a line in a fixed direction from O called the **initial line**.



- The polar coordinates of a point P are written as (r, θ) , where r is the distance of P from O and θ is the angle between OP and the initial line.

- Examples of polar coordinates of points are shown below,

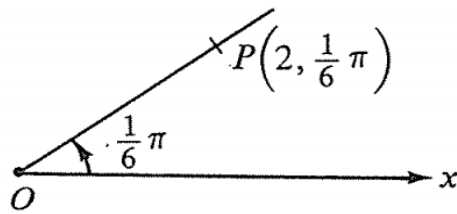


Fig. 6.2

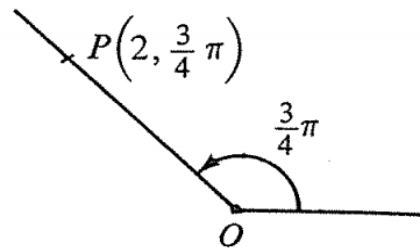
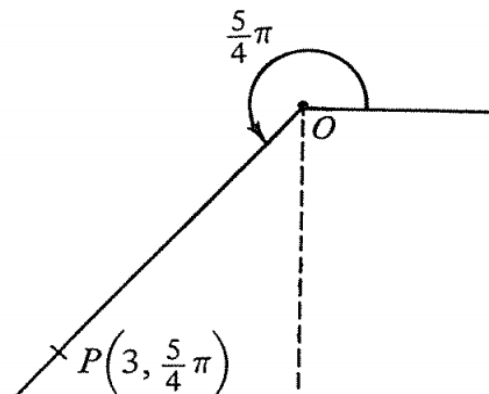
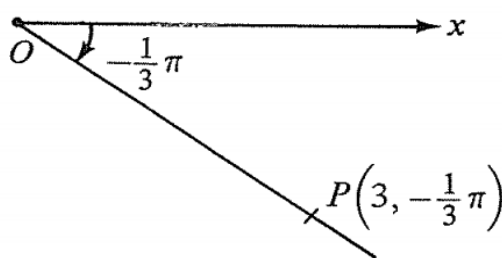
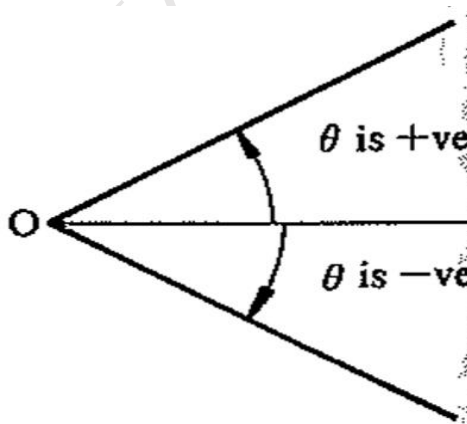


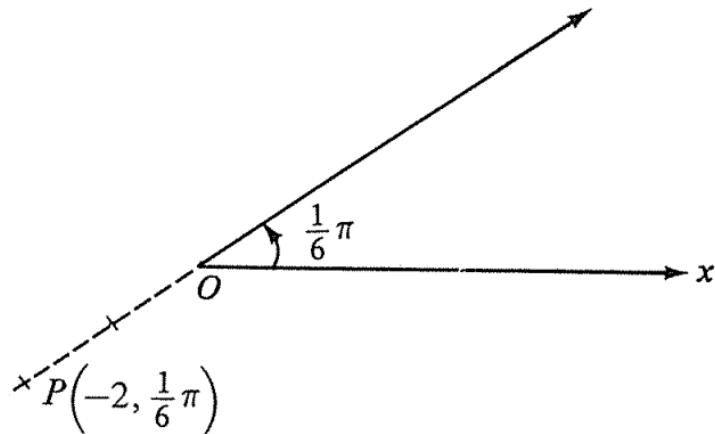
Fig. 6.3



- Positive values of θ correspond to **anti-clockwise** sense of rotation from the initial line, and negative values of θ correspond to clockwise rotation.



- A positive value of r is given to r when the distance is measured in the direction OP , where OP is inclined at an angle θ to the initial line. A negative value of r indicates a distance along PO produced.



- Each point (x, y) in cartesian coordinate system has a unique representation. But in polar coordinates system, $P(r, \theta)$ can be represented in multiple ways.
- For instance, the coordinates of a point $P\left(-2, \frac{\pi}{6}\right)$ can be written as $\left(2, -\frac{5\pi}{6}\right)$ or $\left(2, \frac{7\pi}{6}\right)$ and so on. In order to avoid this multiplicity of forms, it is conventional to give the polar coordinates of a point in the form where $r \geq 0$ and $0 \leq \theta \leq 2\pi$, or $-\pi < \theta \leq \pi$.

The Relation between Cartesian and Polar Coordinates

- If a point P has the cartesian coordinates (x, y) relative to an origin O and polar coordinates (r, θ) relative to pole O and initial line Ox , then

$$x = r \cos \theta, y = r \sin \theta \quad \text{OR} \quad r^2 = x^2 + y^2, \tan \theta = \frac{y}{x}$$

Coordinates Conversion

- **Example 1 (Polar to Cartesian):**

(i) $\left(2, \frac{\pi}{3}\right)$

(ii) $\left(4, -\frac{\pi}{3}\right)$

- **Example 2 (Cartesian to Polar):**

(i) $(1, \sqrt{3})$

(ii) $(2, -2\sqrt{3})$

Equations Conversion**- Example 3 (Polar to Cartesian):**

(i) $r^2(1 + 15 \cos^2 \theta) = 16$

(ii) $r = a \cos 3\theta.$

(iii) $r = (\sqrt{2}) \sec\left(\theta - \frac{1}{4}\pi\right).$

- **Example 4 (Cartesian to Polar):**

(i) $(x^2 + y^2)^2 = a^2(x^2 - y^2)$, where a is a positive constant.

(ii) $y^2 = (x + 1)^2(3 - x)$

(iii) $y^2 - 8x - 16 = 0$

- **Example 5:**

A graph has polar equation $r = \frac{1}{\sin \theta \cos \alpha + \cos \theta \sin \alpha}$, where α is a constant. Hence sketch the graph in the case $\alpha = \frac{\pi}{4}$.

Exercise A:

1. Find the polar equation of the curve whose cartesian equation is represented by

$$y^2 = x^2 \left(\frac{1 - x^2}{1 + x^2} \right).$$

2. Obtain the cartesian equation of the curve whose equation in polar coordinates is

$$r^2(1 + 24 \sin^2 \theta) = 25a^2, 0 \leq \theta < 2\pi,$$

where a is a positive constant.

3. The cartesian equation of a curve is $(x^2 + y^2)^2 = 2a^2xy$, where a is a positive constant. Convert this to polar form.



Polar Curve Tracing

- An equation whose variables are polar coordinates is called a polar equation.
 - The shape of a curve can be determined from its polar equation by listing corresponding values of θ and r and plotting these coordinates. (Values of θ are usually limited to the range $0 \leq \theta < 2\pi$ or $-\pi < \theta \leq \pi$ or a subset of either of these intervals.) Certain observations can reduce the amount of tabulated work however,
- (1) If $r = 0$ when $\theta = \alpha$, then $\theta = \alpha$ is a tangent to the curve at the pole O .
 - (2) If $r = 0$ when $\theta = \alpha$ and $\theta = \beta$, where $\beta > \alpha$, then the curve has a loop between $\theta = \alpha$ and $\theta = \beta$.
 - (3) Look for symmetry if we want to sketch the whole curve.

Polar Axis Symmetry	Line $\theta = \frac{\pi}{2}$ Symmetry	Pole Symmetry

- (4) Write the least and greatest value of r if possible. Otherwise we need to use $\frac{dr}{d\theta} = 0$ to find the least/greatest value of r if the question requires.

Note: The three tests for symmetry are *sufficient* conditions for symmetry, but they are not *necessary* conditions. That is, an equation may fail these tests and still have that specific symmetry. A failure of symmetric means you cannot use that specific symmetry for a shortcut in graphing.

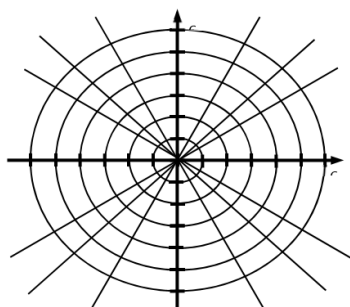


Common Polar Lines and Curves

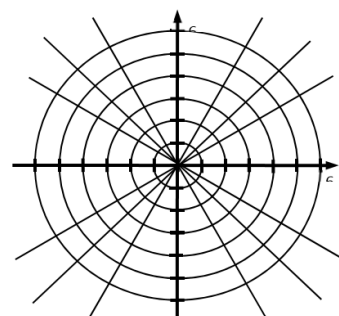
(A) Lines

- If $\theta = \alpha$, we will have equation of line $y = mx$. It is because $\tan \theta = \frac{y}{x}$, so $m = \frac{y}{x}$.
- In fact, if we restrict r to positive values only, then $\theta = \alpha$ describes a 'half-line'.
- The graph of the equation $r \sin \theta = a$ or $r = a \operatorname{cosec} \theta$, is a **horizontal line**. a units above the pole if $a > 0$; $|a|$ units below the pole if $a < 0$, where a is a non-zero real number.
- The graph of the equation $r \cos \theta = a$ or $r = a \sec \theta$, is a **vertical line**. a units to the right of the pole if $a > 0$; $|a|$ units to the left of the pole if $a < 0$, where a is a non-zero real number.
- **Example 6:** Identify and graph each equation.

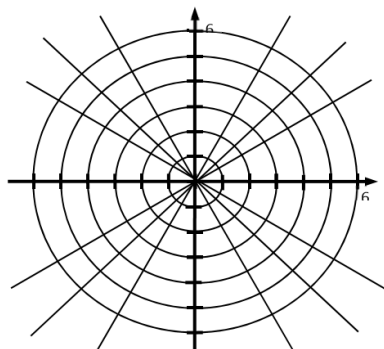
a) $r = 4$



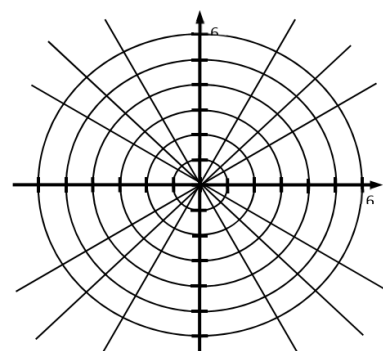
b) $\theta = \frac{3\pi}{4}$



c) $r \sin \theta = 3$



d) $r \cos \theta = -2$





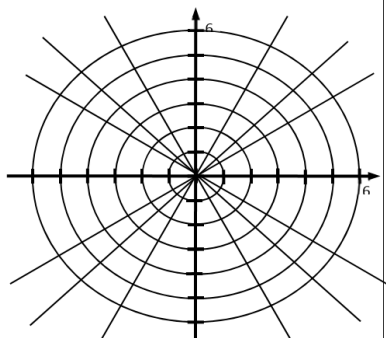
(B) Circles

Let a be a positive constant,

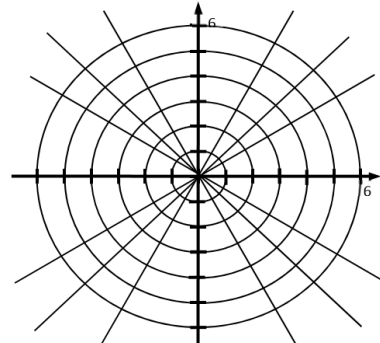
Equations	Description
1. $r = a$	Circle with centre $(0,0)$, radius a .
2. $r = 2a \sin \theta$	Circle with centre $(0,a)$, radius a .
3. $r = -2a \sin \theta$	Circle with centre $(0,-a)$, radius a .
4. $r = 2a \cos \theta$	Circle with centre $(a,0)$, radius a .
5. $r = -2a \cos \theta$	Circle with centre $(-a,0)$, radius a .

- **Example 7:** Identify and graph each equation.

a) $r = 4 \cos \theta$



b) $r = -6 \sin \theta$



(C) Spiral

- Spiral shape will be obtained if polar equation is in the form $r = a\theta$, $r = e^{a\theta}$ or $r = a \ln \theta$, where a is non-zero constant.

- **Example 8 (PYQ M/J 2009/1/5):**

Sketch $r = \theta$, $0 \leq \theta \leq \frac{\pi}{2}$.

- **Example 9 (PYQ M/J 2010/11/12/2):**

Sketch $r = e^{\theta}$, $0 \leq \theta \leq \frac{\pi}{2}$.

- **Example 10 (PYQ M/J 2010/11/12/2):**

Sketch $r = a(1 - e^{-\theta}), 0 \leq \theta \leq 2\pi$.

- **Example 11 (PYQ O/N 2006/1/7):**

Sketch $r = 10 \ln(1 + \theta), 0 \leq \theta \leq \frac{\pi}{2}$.



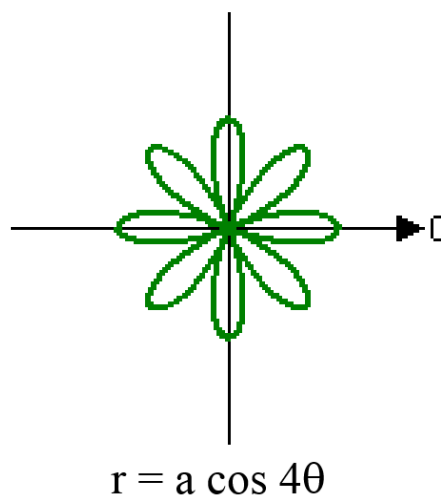
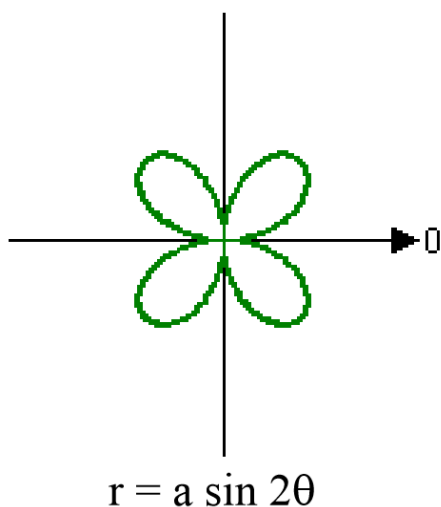
(D) Rose

- A rose curve is a graph that is produced from a polar equation in the form of

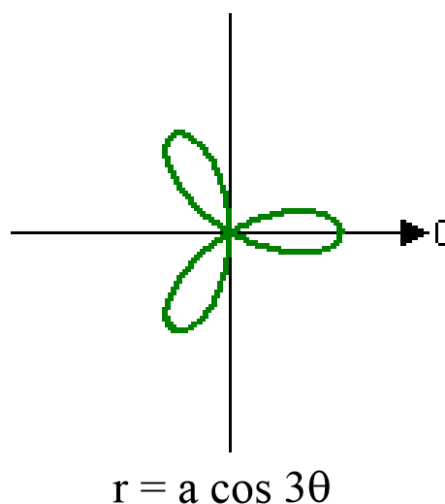
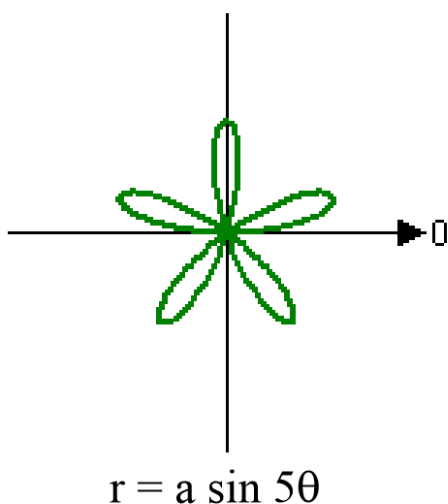
$$r = a \sin n\theta \quad \text{or} \quad r = a \cos n\theta$$

where $a \neq 0$ and n is an integer > 1 .

- They are called rose curves because the loops that are form resemble petals. The number of petals that are present will depends on the value of n . The value of a will determine the length of the petals.
- If n is an even integer, then the rose will have $2n$ petals.



- If n is an odd integer, then the rose will have n petals.



- **Example 12 (PYQ M/J 2011/11/12/2):**

Sketch $r = 2a \cos 2\theta, 0 \leq \theta \leq \frac{\pi}{4}$.

- **Example 13:**

Sketch $r = a \cos 3\theta, 0 \leq \theta \leq 2\pi$.



- **Example 14:**

Sketch $r = a \sin^2 2\theta, 0 \leq \theta \leq 2\pi$.

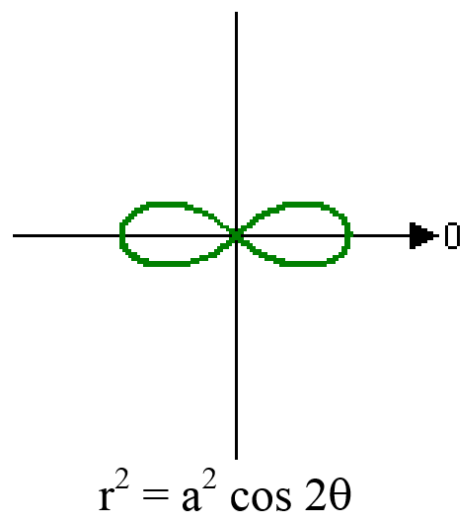
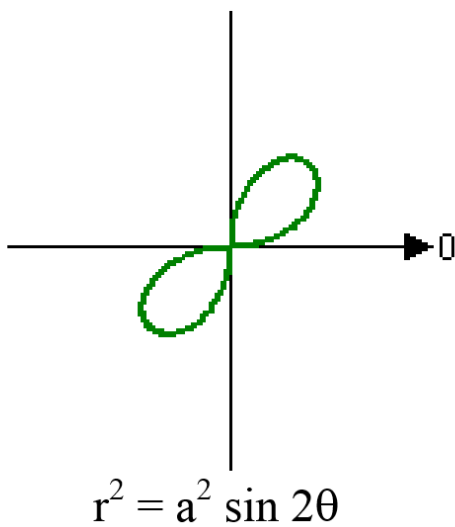
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**(E) Lemniscates**

- Lemniscates has the shape of a figure-8 or a propeller. Lemniscates have the general polar equation of

$$r^2 = a^2 \sin 2\theta \quad \text{or} \quad r^2 = a^2 \cos 2\theta, \text{ where } a \neq 0.$$

- A lemniscate containing the sine function will be symmetric to the pole.
- A lemniscate containing the cosine function will be symmetric to the polar axis, to $\theta = \frac{\pi}{2}$, and the pole.





- **Example 15 (PYQ M/J 2014/13/4):**

Sketch $r^2 = a^2 \sin 2\theta$, $-\pi \leq \theta \leq \pi$.

- **Example 16 (PYQ M/J 2012/13/11E):**

Sketch $r^2 = a^2 \cos 2\theta$, $-\pi \leq \theta \leq \pi$.

(F) Cardioid

- Cardioid has a heart-shaped curve. Cardioid have the polar equation in the form

$$r = a(1 + \cos \theta) \quad \text{or} \quad r = a(1 - \cos \theta)$$

$$r = a(1 + \sin \theta) \quad \text{or} \quad r = a(1 - \sin \theta)$$



- **Example 17 (PYQ O/N 2017/11/12/13):**

Sketch $r = a + a \cos \theta, 0 \leq \theta \leq 2\pi$.

- **Example 18 (PYQ M/J 2014/11/12/5):**

Sketch $r = a + a \sin \theta, 0 \leq \theta \leq 2\pi$.

(G) Limacons

- Limacons has a snail-shaped curve. Limacons have the polar equation in the form

$$r = a \pm b \sin \theta \quad \text{or} \quad r = a \pm b \cos \theta, \text{ where } a, b > 0$$

- The limacons containing **sine** will be *above the horizontal axis* if the sign between a and b is (+), or *below the horizontal axis* if the sign is (-).
- The limacons containing **cosine** will be the *right of the vertical axis* if the sign between a and b is (+), or the *left of the vertical axis* if the sign is (-).
- The ratio of $\frac{a}{b}$ will determine the exact shape of the limacons.

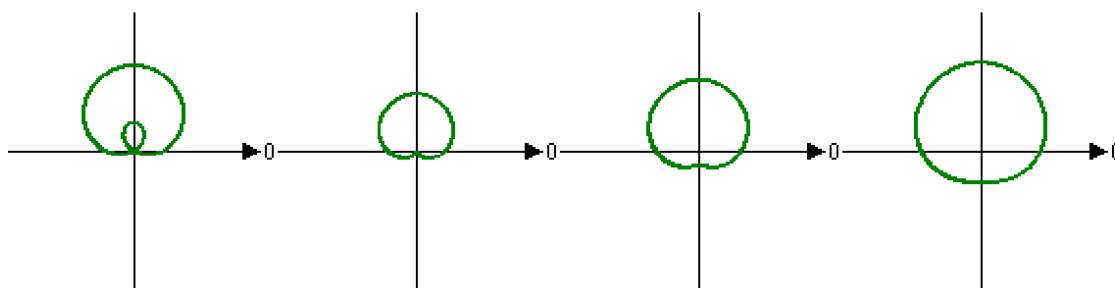
$$\frac{a}{b} < 1$$

$$\frac{a}{b} = 1$$

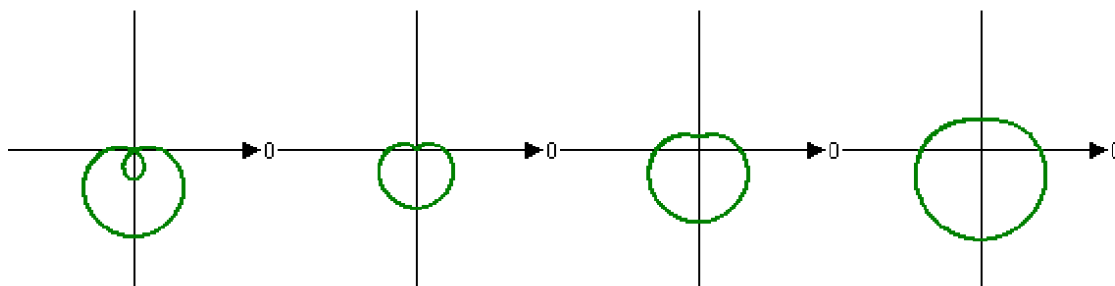
$$1 < \frac{a}{b} < 2$$

$$\frac{a}{b} \geq 2$$

$$r = a + b \sin \theta$$



$$r = a - b \sin \theta$$



- The graphs of limacons with cosine would have similar shapes but along the horizontal axis.

- **Example 19 (PYQ O/N 2011/13/8):**

Sketch $r = 3 + 2 \cos \theta$, $-\pi \leq \theta \leq \pi$.

- **Example 20:**

Sketch $r = 1 + 2 \cos \theta$, $-\frac{2}{3}\pi \leq \theta \leq \frac{2}{3}\pi$.

Exercise B:

Sketch each of the following curves:

(a) $r = e^{2\theta} - e^{-2\theta}, 0 \leq \theta \leq \frac{\pi}{2}$. (PYQ O/N 2019/11/12/13/11O)

(b) $r^2 = \ln(1 + \theta), 0 \leq \theta \leq 2\pi$. (PYQ M/J 2019/13/2)

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(c) $r = 2a|\cos \theta|$, $0 \leq \theta \leq \pi$. (PYQ M/J 2018/13/8)

(d) $r = -2a \sin 2\theta$, $0 \leq \theta \leq 2\pi$. (PYQ M/J 2017/13/11E)

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(e) $r = \sqrt{\sin \frac{1}{2}\theta}$, $0 \leq \theta \leq \pi$. (PYQ M/J 2015/11/12/5)

(f) $r = 1 + \sin \theta$, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. (PYQ O/N 2011/13/8)

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(g) $r = a \sin 3\theta$, $0 \leq \theta \leq \frac{\pi}{3}$. (PYQ O/N 2009/1/10)

(h) $r = \left(\frac{\pi}{2} - \theta\right)^2$, $0 \leq \theta \leq \frac{\pi}{2}$. (PYQ O/N 2008/1/3)

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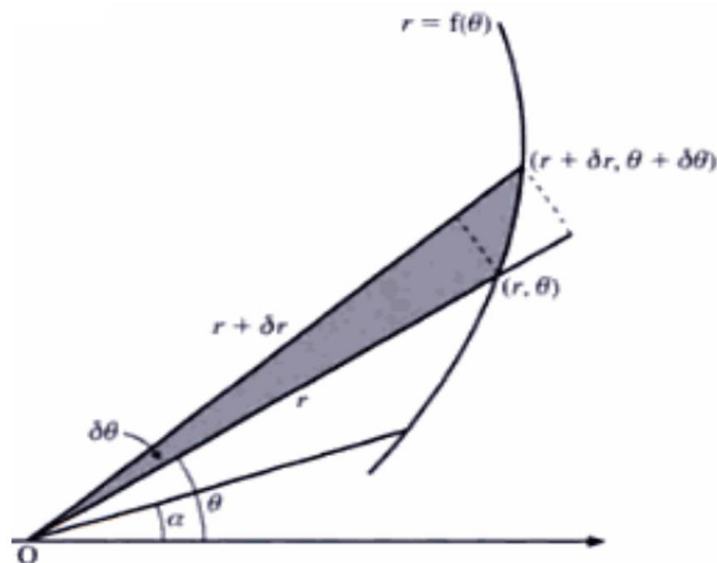


(i) $r = \theta \sin \theta$, $0 \leq \theta \leq \pi$. (PYQ O/N 2007/1/7)

(j) $r = \frac{\pi - \theta}{\theta}$, $\frac{1}{2}\pi \leq \theta \leq \pi$. (PYQ M/J 2005/1/6)

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Area of a Sector in Polar Coordinates



- Let A be the area bounded by the curve $r = f(\theta)$ and the two radii at α and at θ .
- As θ increases by $\delta\theta$, the increase in area, δA , shown shaded, using areas of sectors, is given by

$$\frac{1}{2}r^2\delta\theta \leq \delta A \leq \frac{1}{2}(r + \delta r)^2\delta\theta$$

- Dividing throughout by $\delta\theta$, we obtain

$$\frac{1}{2}r^2 \leq \frac{\delta A}{\delta\theta} \leq \frac{1}{2}(r + \delta r)^2$$

- As $\delta\theta \rightarrow 0$, $\frac{\delta A}{\delta\theta} \rightarrow \frac{dA}{d\theta}$ and $\delta r \rightarrow 0$. Therefore, we have

$$\frac{dA}{d\theta} = \frac{1}{2}r^2$$

- Integrating both sides with respect to θ , we obtain

$$\int \frac{dA}{d\theta} d\theta = \frac{1}{2} \int r^2 d\theta$$

$$A = \frac{1}{2} \int r^2 d\theta$$

- Therefore, the general equation for the area of a sector of a curve is

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

when the area is bounded by the radii $\theta = \alpha$ and $\theta = \beta$.



- **Example 21:**

Find the area of $r = a(1 + \cos \theta)$.

- **Example 22:**

Find the area of one loop of the curve $r = a \cos 3\theta$.

- **Example 23:**

Find the area enclosed by the curve $r = a \cos 2\theta$.

- **Example 24:**

Find the area enclosed by the curve $r^2 = a^2 \sin 2\theta$.

- **Example 25:**

Sketch the curves C_1 and C_2 whose respective polar equations are

$$C_1: r = 1 + 2 \cos \theta, \quad 0 \leq \theta \leq \frac{2}{3}\pi,$$

$$C_2: r = 1 + \cos \theta, \quad 0 \leq \theta < 2\pi.$$

Find the area of the finite region bounded by C_1 and C_2 and the half lines $\theta = 0$ and $\theta = \frac{\pi}{2}$.

- **Example 26:**

The curves C_1 and C_2 have polar equations

$$C_1: r = 4 \sin^2 \theta, \quad 0 \leq \theta \leq 2\pi,$$

$$C_2: r = 2\sqrt{3} \sin 2\theta, \quad 0 \leq \theta < \frac{\pi}{2}.$$

- (i) Sketch C_1 and C_2 on the same diagram.
- (ii) Find the polar coordinates of all points of intersection of C_1 and C_2 .
- (iii) Find, to 2 decimal places, the area of the region R which is inside both C_1 and C_2 .

- **Example 27 (PYQ M/J 2013/13/10):**

Use the identity $2 \sin P \cos Q \equiv \sin(P + Q) + \sin(P - Q)$ to show that

$$2 \sin \theta \cos\left(\theta - \frac{1}{4}\pi\right) \equiv \cos\left(2\theta - \frac{3}{4}\pi\right) + \frac{1}{\sqrt{2}}. \quad [3]$$

A curve has polar equation $r = 2 \sin \theta \cos\left(\theta - \frac{1}{4}\pi\right)$, for $0 \leq \theta \leq \frac{3}{4}\pi$. Sketch the curve and state the polar equation of its line of symmetry, justifying your answer. [3]

Show that the area of the region enclosed by the curve is $\frac{3}{8}(\pi + 1)$. [6]

- **Example 28 (PYQ O/N 2013/13/10):**

The curve C has polar equation $r = 2 \sin \theta (1 - \cos \theta)$, for $0 \leq \theta \leq \pi$. Find $\frac{dr}{d\theta}$ and hence find the polar coordinates of the point of C that is furthest from the pole. [5]

Sketch C . [2]

Find the exact area of the sector from $\theta = 0$ to $\theta = \frac{1}{4}\pi$. [6]

- **Example 29 (PYQ M/J 2012/11/12/4):**

The curve C has polar equation $r = 2 + 2 \cos \theta$, for $0 \leq \theta \leq \pi$. Sketch the graph of C . [2]

Find the area of the region R enclosed by C and the initial line. [4]

The half-line $\theta = \frac{1}{5}\pi$ divides R into two parts. Find the area of each part, correct to 3 decimal places. [3]

- **Example 30 (PYQ M/J 2009):**

- 5 Draw a sketch of the curve C whose polar equation is $r = \theta$, for $0 \leq \theta \leq \frac{1}{2}\pi$. [2]

On the same diagram draw the line $\theta = \alpha$, where $0 < \alpha < \frac{1}{2}\pi$. [1]

The region bounded by C and the line $\theta = \frac{1}{2}\pi$ is denoted by R . Find the exact value of α for which the line $\theta = \alpha$ divides R into two regions of equal area. [4]

- **Example 31 (PYQ M/J 2010/13/11):**

- 11** The curve C has polar equation

$$r = \frac{a}{1 + \theta},$$

where a is a positive constant and $0 \leq \theta \leq \frac{1}{2}\pi$.

- (i) Show that r decreases as θ increases. [2]

- (ii) The point P of C is further from the initial line than any other point of C . Show that, at P ,

$$\tan \theta = 1 + \theta,$$

and verify that this equation has a root between 1.1 and 1.2. [4]

- (iii) Draw a sketch of C . [3]

- (iv) Find the area of the region bounded by the initial line, the line $\theta = \frac{1}{2}\pi$ and C , leaving your answer in terms of π and a . [3]

- **Example 32 (PYQ O/N 2009):**

- 10 The curve C has polar equation

$$r = a \sin 3\theta,$$

where $0 \leq \theta \leq \frac{1}{3}\pi$.

- (i) Show that the area of the region enclosed by C is $\frac{1}{12}\pi a^2$. [3]

- (ii) Show that, at the point of C at maximum distance from the initial line,

$$\tan 3\theta + 3 \tan \theta = 0. \quad [3]$$

- (iii) Use the formula

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

to find this maximum distance. [4]

- (iv) Draw a sketch of C . [2]

BONUS: To Sketch a Polar Curve with Asymptotes**- Example 33:**

Sketch $r = a \tan \theta, 0 \leq \theta \leq \pi$.

- Example 34:

Sketch $r = a(2 \sec \theta - 1), -\pi \leq \theta \leq \pi$.

- **Example 35:**

Show that, in polar coordinates, the equation $r = \operatorname{cosec} \theta$, $0 < \theta < \pi$ represent a straight line and show this line in a sketch.

Hence, or otherwise, sketch the curve whose polar equation is

$$r = \operatorname{cosec} \theta - 2, 0 < \theta < \pi.$$

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- **Example 36 (PYQ O/N 2002):**

The curve C has polar equation $r\theta = 1$, for $0 < \theta \leq 2\pi$.

- (i) Use the fact that $\frac{\sin \theta}{\theta}$ tends to 1 as θ tends to 0 to show that the line with cartesian equation $y = 1$ is an asymptote to C . [2]

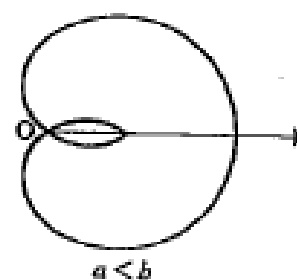
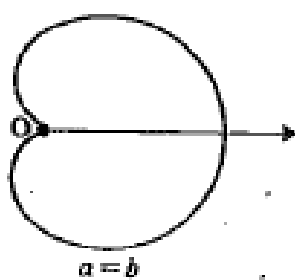
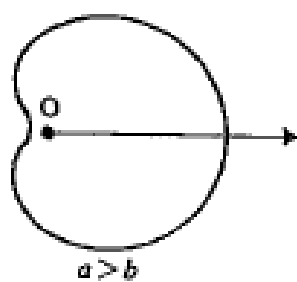
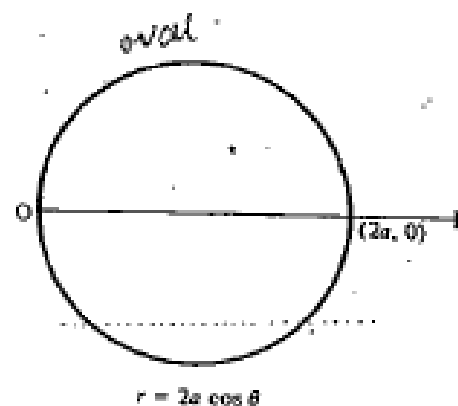
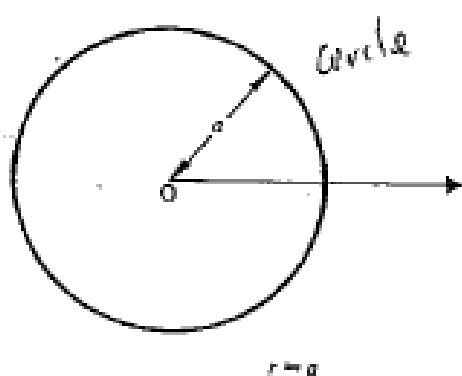
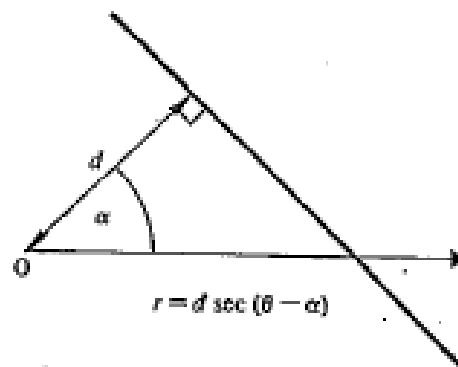
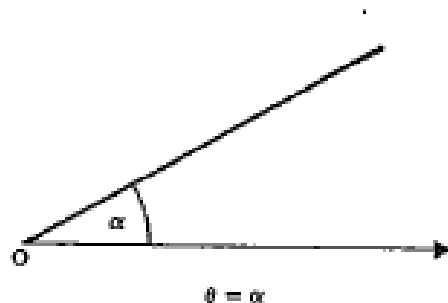
- (ii) Sketch C . [1]

The points P and Q on C correspond to $\theta = \frac{1}{6}\pi$ and $\theta = \frac{1}{3}\pi$ respectively.

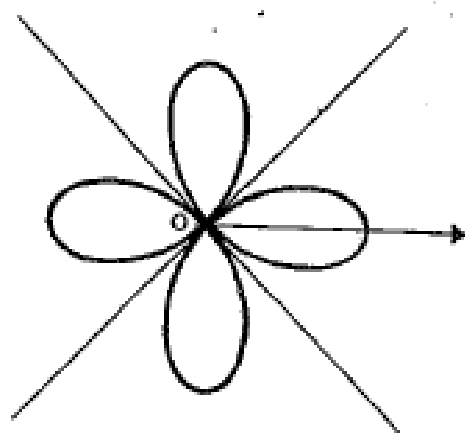
- (iii) Find the area of the sector OPQ , where O is the origin. [3]

Summary:

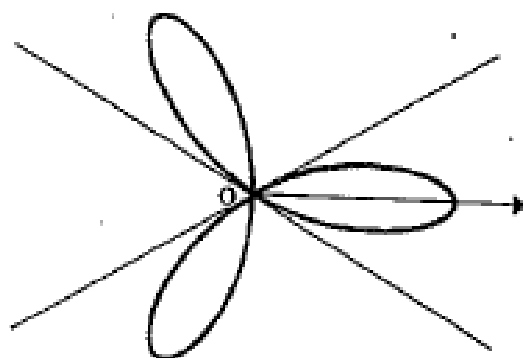
COMMON POLAR LINES AND CURVES



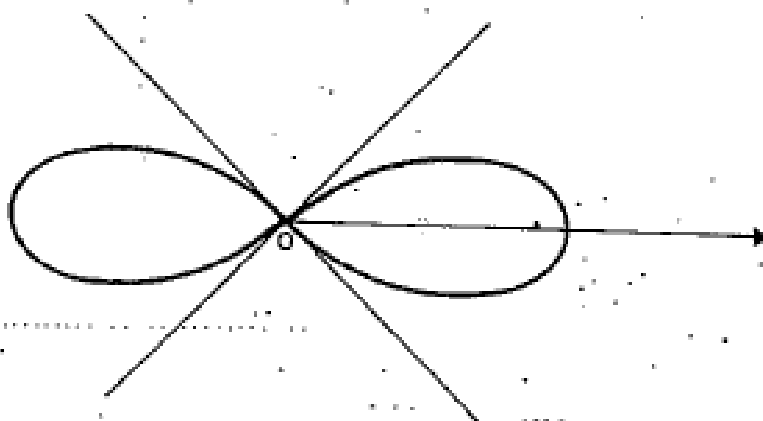
$r = a + b \cos \theta$



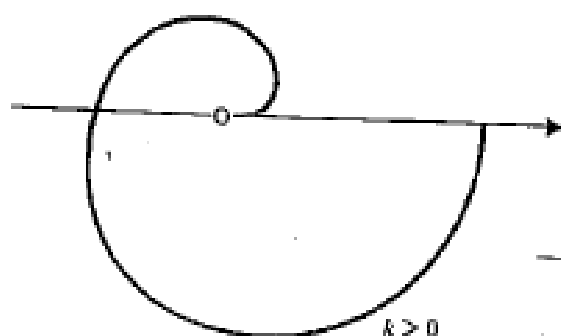
$$r = a \cos 2\theta$$



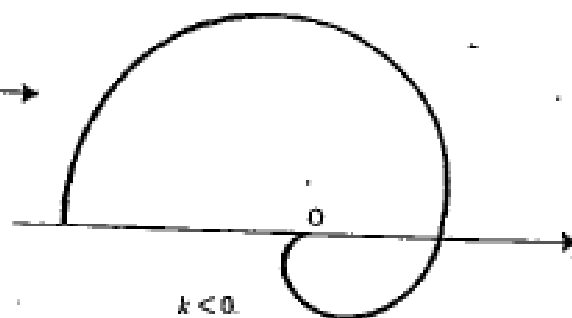
$$r = a \cos 3\theta$$



$$r^2 = a^2 \cos 2\theta$$



$$k > 0$$

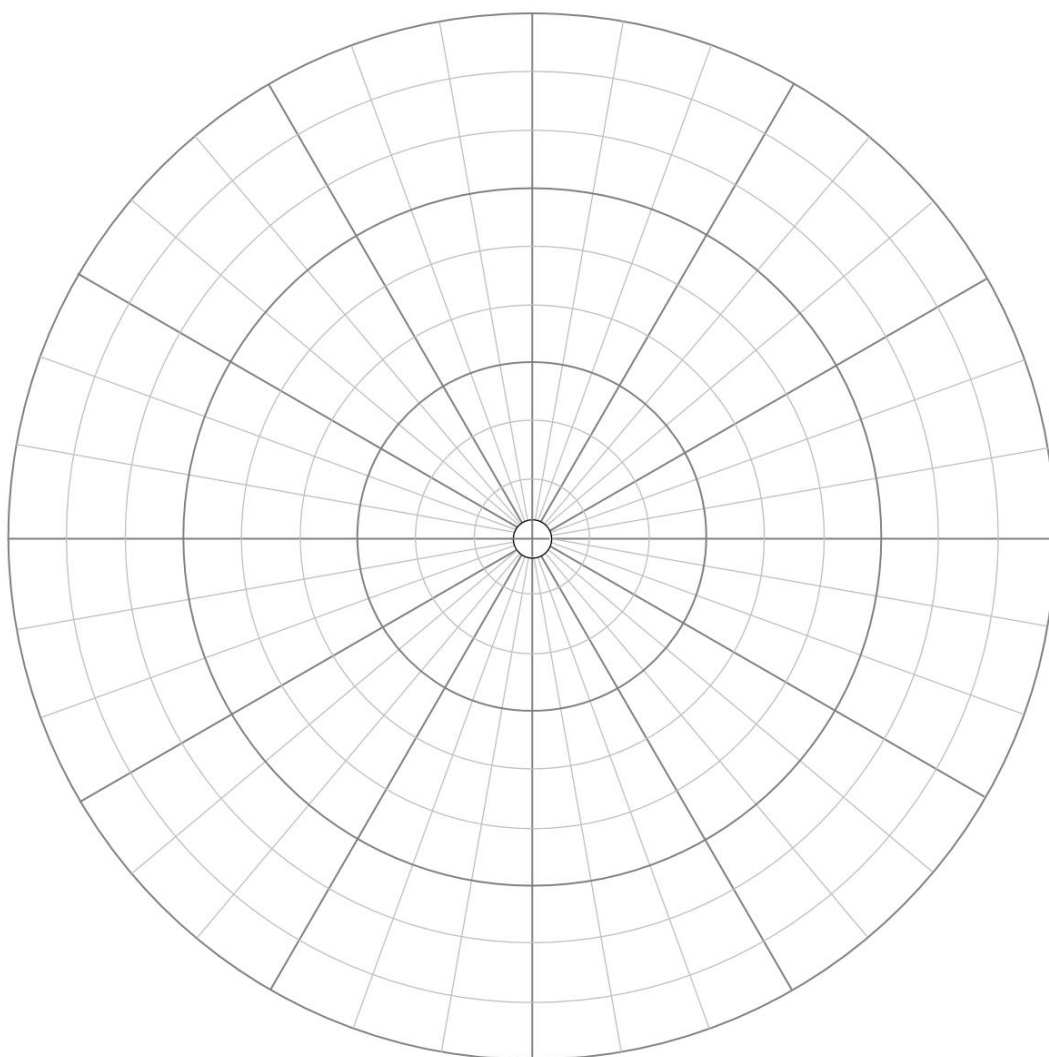


$$k < 0$$

$$r = k\theta$$



Appendix: Polar Coordinate Graph Paper



Sunway